



**FACULTY OF SCIENCE, ENGINEERING AND
COMPUTING**

SCHOOL OF COMPUTER SCIENCE AND MATHEMATICS

**BSc (HONS) DEGREE IN
CYBER SECURITY & DIGITAL FORENSICS**

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K2121774

Investigating Cyber Psychology In the Context of User
Behaviour and Email-Based Malware

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Declaration:

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Abstract

Phishing attacks remain one of the most pervasive cybersecurity threats, leveraging psychological manipulation to exploit users' cognitive biases and human error. Despite technological advancements in email security, phishing attacks continue to increase in both frequency and sophistication, underscoring the critical need for targeted awareness interventions. This study investigates cyber psychology's role in user behaviour concerning email-based malware and phishing susceptibility.

The research employs a structured approach, integrating pre- and post-test quizzes to assess users' ability to identify phishing emails before and after an educational intervention. By focusing on cognitive biases, decision-making heuristics, and behavioural tendencies, the study aims to enhance phishing awareness training effectiveness. The findings will contribute to refining cybersecurity education, ensuring more resilient defences against email-based social engineering attacks.

All references and citations adhere to the Harvard referencing style, ensuring proper academic rigour and alignment with university guidelines.

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Glossary of Terms

- **Phishing:** A cyber-attack that attempts to deceive users into providing sensitive information by pretending to be a trustworthy entity.
- **Cyber Psychology:** The study of human behaviour in digital environments, particularly in relation to cybersecurity threats.
- **Social Engineering:** Manipulative tactics used by cybercriminals to deceive users into divulging confidential information.
- **Decision Fatigue:** A cognitive state in which individuals struggle to make rational decisions due to excessive information processing.
- **Security Fatigue:** A phenomenon where users become desensitised to security warnings and best practices due to overexposure.
- **Cognitive Biases:** Systematic patterns of deviation from rationality in judgment, affecting decision-making in phishing scenarios.
- **Authentication Mechanisms:** Security measures, such as multi-factor authentication, used to verify user identities.
- **Machine Learning-Driven Threat Detection:** AI-powered techniques used to identify phishing emails based on patterns and anomalies.
- **Human Error:** Mistakes made by users, often due to lack of awareness, leading to security vulnerabilities.

1. Introduction

1.1 Context and Relevance

Phishing attacks are among the most persistent cyber threats, affecting individuals, businesses, and institutions alike. As digital communication continues to grow, email remains a primary channel for both professional and personal exchanges, making it an attractive target for cybercriminals. The success of phishing schemes is driven by the manipulation of human psychology—attackers exploit deception, urgency, and trust to trick individuals into disclosing sensitive information.

The field of cyberpsychology is becoming increasingly important in phishing awareness as it seeks to understand human cognitive behaviours that lead to cybersecurity vulnerabilities. Research indicates that cognitive biases, decision fatigue, and habitual clicking behaviours significantly increase the likelihood of falling for phishing attacks. By studying these behavioural aspects, cybersecurity training can be tailored to address these vulnerabilities and improve user awareness.

1.2 Problem Statement

Despite advancements in email filtering, authentication protocols, and AI-driven threat detection, phishing remains the most effective form of cyberattack. According to the Anti-Phishing Working Group (APWG, 2024), there were 932,923 recorded phishing attacks globally in Q3 2024—marking a significant increase from previous years. Furthermore, phishing-related cybercrime costs organizations an estimated £12.4 billion annually, with an average financial loss of £1,200 per affected employee (Paubox, 2024).

Phishing accounts for over 90% of cybersecurity breaches, with 83% of businesses reporting at least one phishing-related incident in the past year (AAG IT, 2024). These statistics emphasize the persistent threat phishing poses and indicate that traditional security measures alone are not enough. Attackers continually refine their strategies, leveraging social engineering tactics that exploit emotions such as fear, anticipation of rewards, and urgency.

Another major factor contributing to phishing vulnerability is the cognitive overload experienced by users who receive large volumes of emails daily. Research shows that decision fatigue and security fatigue impair users' ability to assess emails critically (Ge et al., 2021). Without incorporating cyberpsychology principles into phishing awareness programs, users remain highly susceptible to these deceptive tactics.

1.3 Research Aim and Objectives

Research Aim

This study aims to examine the role of cyberpsychology in phishing susceptibility and develop an intervention framework that improves phishing detection through behavioural insights.

Research Objectives

- Identify key psychological traits that influence phishing susceptibility.
- Evaluate the effectiveness of existing phishing awareness programs in addressing cognitive biases.
- Develop an interactive phishing education framework that incorporates behavioural insights.
- Implement and assess the effectiveness of a training intervention using pre- and post-test evaluations.
- Measure the impact of the intervention on phishing detection rates.
- Provide recommendations for enhancing cybersecurity training based on behavioural insights.

1.4 Scope of the Project

This research focuses specifically on phishing attacks conducted via email, excluding other social engineering methods such as vishing (voice phishing) and smishing (SMS phishing). Email phishing is the primary focus due to its high prevalence and effectiveness in cybercrime, making it a leading entry point for attackers.

The study will examine individuals in corporate and academic environments, as these groups frequently interact with emails and are common targets of phishing attacks. Corporate employees are particularly vulnerable due to their handling of sensitive financial and business-related data, while academic users often manage research data and institutional credentials.

The intervention framework will incorporate behavioural training strategies, including interactive phishing simulations and psychology-based awareness techniques, to enhance phishing resilience. This study will not focus on highly technical anti-phishing solutions (e.g., email filtering algorithms) but will instead address the human factor, which remains the weakest link in cybersecurity.

Additionally, due to time constraints and accessibility limitations, the project will emphasize phishing education rather than technical development. The findings will contribute to the

advancement of cybersecurity awareness strategies that can be widely applied across various industries.

1.5 Ethical, Legal, and Social Considerations

Ethical Considerations

This research involves human participants who will complete phishing awareness assessments. To ensure ethical compliance:

- Informed consent will be obtained from all participants before they engage in phishing simulation exercises.
- No personally identifiable information will be collected, ensuring participant anonymity (Bada, Sasse, & Nurse, 2019).
- Participants will have the right to withdraw at any stage without any repercussions.

As the study does not involve deception or data manipulation, it is expected to meet ethical standards without requiring extensive institutional ethical approval.

Legal Considerations

The study will adhere to UK GDPR regulations, ensuring that all participant data is:

- Securely stored and used solely for research purposes.
- Not shared with third parties or used beyond the scope of the study (López-Aguilar & Solanas, 2021).
- Additionally, the project aligns with cybersecurity laws and regulations, ensuring that phishing simulations comply with ethical hacking guidelines.

Social Considerations

This research aims to positively impact cybersecurity awareness, benefiting both individuals and organizations. Key considerations include:

- Empowering users by equipping them with knowledge to identify and mitigate phishing threats.
- Avoiding fear-based tactics and instead focusing on educational and psychology-driven training (Vishwanath et al., 2018).
- Contributing to organizational security strategies, particularly in industries highly targeted by phishing attacks, such as finance, education, and healthcare.

2. Project Management Strategy

2.1 Introduction to Project Management Strategy

Effective project management is essential for the successful execution of this research project. The structured management of research, planning, implementation, and documentation phases ensures smooth progress and timely completion. The project initially followed a structured Waterfall approach, where each phase was planned sequentially. However, due to coursework constraints, adjustments were made to reflect a more adaptive approach. This section outlines the project management methodologies, tools, and techniques used to efficiently manage time, research, and deliverables.

Key constraints considered in this project included:

- Time Management: Ensuring the research and implementation were completed within the academic deadlines.
- Resource Management: Securing access to essential research materials, software, and guidance from supervisors.
- Scope Management: Maintaining a strict focus on phishing susceptibility and cyber psychology, without expanding into unrelated cybersecurity areas.

A structured approach, incorporating project management tools such as a Gantt chart (Figure 1), a simple checklist (Figure 2), and risk assessment techniques, facilitated the seamless execution of tasks.

2.2 Tools and Techniques Used

Planning and Scheduling

- Gantt Chart (Figure 1)

A Gantt chart was initially created to outline key milestones, deadlines, and task dependencies. However, the original timeline was overly optimistic, underestimating coursework requirements and unforeseen delays. Consequently, a revised Gantt chart was developed to present a more realistic schedule that accommodated academic responsibilities.

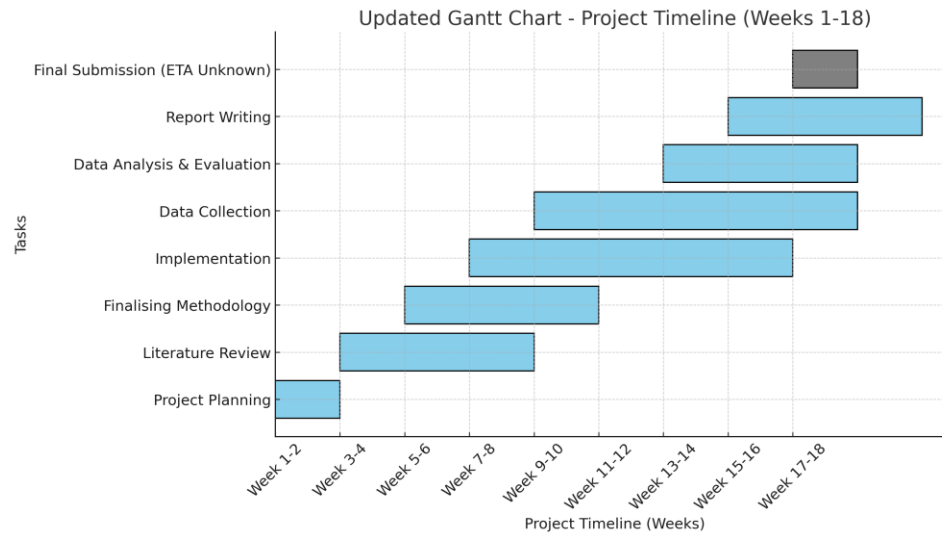


Figure 1: Gantt Chart

- Simple Checklist (Figure 2)

A simple checklist was implemented to track research progress and milestones. Tasks were categorized as follows:

- Pending: Tasks yet to be started.
- Ongoing: Tasks currently in progress.
- Completed: Finalized components.

Task	Status
Research Collection	Completed
Methodology Planning	Completed
Introduction	Completed
Project Management Strategy	Completed
Literature Review	Ongoing
Data Collection	Pending
Data Analysis	Pending
Draft Writing	Ongoing
Citations/Sources	Ongoing
Final Report Review	Pending ¹

Figure 2: Simple Checklist

This approach facilitated systematic tracking of critical project elements without necessitating the use of advanced project management software.

Documentation and Research Management

- Microsoft Word: Utilized for writing and structuring the research report.
- Zotero/Mendeley: Reference management tools used for storing, organizing, and formatting citations in Harvard referencing style.
- Excel/Google Sheets: Used for data collection, survey responses, and phishing susceptibility analysis.

Risk Management and Mitigation

- SWOT Analysis (Figure 3): Conducted to evaluate the strengths, weaknesses, opportunities, and threats associated with the project.

Category	Details
Strengths	Clear research focus, access to academic materials.
Weaknesses	Time constraints, difficulty in gathering participant responses.
Opportunities	Contributing to phishing awareness education
Threats	Limited participation in phishing tests, potential software access issues.

Figure 3: SWOT Analysis

- Mitigation Strategies:
 - Research activities were planned to prevent last-minute workload accumulation.
 - Cloud-based storage (e.g., OneDrive) was used to prevent data loss.
 - Project meetings were attended online to facilitate efficient communication with supervisors.
 - The project timeline was adjusted through an updated Gantt chart (Figure 1) to ensure realistic work distribution.

2.3 Alternative Tools and Techniques Considered

Several alternative project management tools were reviewed but not implemented due to feasibility constraints:

- Microsoft Project: More suitable for large-scale technical projects; however, Excel-based Gantt charts were sufficient for this project.
- Monday.com/Asana: Advanced project tracking platforms were considered, but after evaluating Trello, it was determined that a simple checklist approach was more appropriate for managing tasks and milestones. Trello provided structured task tracking, but given the project's scope and straightforward nature, a checklist was more efficient and practical.

2.4 Challenges and Lessons Learned

Challenges Encountered

- Time Constraints: The initial scheduling plan was overly optimistic, necessitating a revised Gantt chart (Figure 1).
- Data Collection Challenges: Gathering participant data for phishing awareness testing was slower than expected.
- Software Limitations: Certain cybersecurity tools required paid access, leading to the use of alternative academic sources.

Lessons Learned

- Early planning and milestone tracking using the Gantt chart (Figure 1) ensured steady project progress.
- The simple checklist approach (Figure 2) enabled systematic task tracking without requiring sophisticated project management software.
- Project online meetings provided valuable guidance and helped refine the project's scope and direction.
- Adapting project timelines was essential, as real-world constraints necessitated modifications to the original schedule.

2.5 Summary

Effective project management was instrumental in ensuring that the research, planning, implementation, and documentation phases were completed within a structured timeframe. Initially, the project followed a Waterfall approach, where each phase was planned in sequence. However, due to coursework demands, the Gantt chart (Figure 1) was revised to

establish a more realistic schedule, demonstrating an adaptive approach to project management.

To monitor progress, a simple checklist (Figure 2) was implemented, categorizing tasks into pending, ongoing, and completed stages. This method provided a clear overview of task progression without requiring advanced project management tools.

Additionally, a SWOT Analysis (Figure 3) was conducted to assess strengths, weaknesses, opportunities, and threats. This analysis facilitated risk anticipation, such as time constraints and software limitations, leading to the development of effective mitigation strategies.

The Gantt chart (Figure 1) played a key role in identifying overlapping tasks and reorganizing the workflow to reduce time pressure. By restructuring project phases, it ensured that research, data collection, and writing were balanced efficiently.

In conclusion, the combination of structured scheduling, simplified task tracking, and risk assessment ensured that project milestones were met efficiently. While alternative tools were considered, they were deemed unnecessary given the project's scope and timeframe. This structured approach contributed to a well-organized and data-driven study on phishing susceptibility and cyber psychology.

3. Literature Review

3.1 Overview of Existing Research

Phishing attacks remain a significant cybersecurity threat, exploiting human psychology to manipulate individuals into divulging sensitive information. Extensive research has been conducted in cyber psychology, phishing susceptibility, and email-based malware, providing valuable insights into how and why users fall victim to phishing scams. This section critically examines existing literature on phishing trends, user vulnerability factors, and training interventions.

This study aligns with previous research on phishing susceptibility, which has identified cognitive biases, decision fatigue, and security overload as significant contributors to user vulnerability (Frauenstein & Flowerday, 2020). Research has further highlighted that technical defences alone are insufficient, as social engineering tactics continue to bypass even the most sophisticated filtering mechanisms (Vishwanath et al., 2018).

3.2 Psychological Factors in Phishing Susceptibility

Cognitive Biases and Decision Fatigue

Studies in cyber psychology suggest that users frequently rely on cognitive shortcuts when evaluating emails, increasing their susceptibility to phishing attacks (Frauenstein & Flowerday, 2020). Key psychological factors include:

- **Authority Bias:** Users are more likely to trust phishing emails that appear to come from authoritative figures such as CEOs or IT administrators.
- **Urgency Effect:** Attackers create a false sense of urgency, pressuring users into making hasty decisions without proper scrutiny (Vishwanath et al., 2018).
- **Habitual Clicking Behaviour:** Users who process large volumes of emails daily often develop automatic clicking habits, reducing their ability to detect fraudulent messages (Sarno et al., 2023).

Security Fatigue and Overconfidence

- **Security fatigue** occurs when users become desensitised to frequent security warnings, increasing their vulnerability to phishing attempts (Tornblad et al., 2021).

- Many users exhibit overconfidence in their ability to detect phishing emails, despite research indicating that even cybersecurity professionals can fall victim to well-crafted phishing attacks (Vishwanath et al., 2016).

3.3 Phishing Trends and Attack Strategies

Phishing techniques have evolved significantly, with attackers leveraging personalised and AI-driven tactics to increase their success rates. Research identifies several common phishing attack strategies:

- Spear Phishing: Highly targeted attacks that use personalised information to deceive individuals.
- Whaling Attacks: Phishing campaigns specifically aimed at high-profile executives within organisations (Rocha et al., 2017).
- Clone Phishing: Attackers replicate legitimate emails and resend them with malicious modifications.

AI in Phishing Attacks and Detection

- AI and automation have enhanced phishing campaigns, with attackers using deep learning models to craft contextually relevant phishing emails that evade traditional detection mechanisms (Sarno et al., 2023).
- *Real-world Example:* Google's Safe Browsing AI detects and blocks over 100 million phishing sites daily, highlighting the increasing role of machine learning in phishing mitigation (Google Security Blog, 2023).
- *Real-world Example:* Microsoft's Defender SmartScreen employs AI-powered phishing detection to identify fraudulent websites and prevent credential theft in Microsoft Edge (Microsoft Security, 2023).

3.4 Review of Phishing Mitigation Systems, Techniques, and Tools

To combat phishing threats, various systems, techniques, and tools have been developed. This section assesses their advantages and limitations using a comparison table (Figure 4) that summarises them:

System/Technique/Tool	Benefits	Limitations
Email Filtering & AI-Based Detection	Automatically blocks suspicious emails, reducing manual workload	Can produce false positives and negatives; attackers constantly adapt to bypass filters
Simulated Phishing Training	Helps users recognise phishing attempts and enhances awareness	Effectiveness depends on frequency and realism; may not fully replicate real-world phishing complexity
Cybersecurity Awareness Programmes	Educates users on phishing indicators, reaching a broad audience	Passive learning may not lead to long-term behavioural changes; lacks adaptability
Multi-Factor Authentication (MFA)	Provides an extra security layer beyond passwords, mitigating account compromise risks	Not foolproof; some users find it inconvenient; phishing attacks can still exploit weaker factors

Figure 4: Comparison Table

3.5 Research Focus and Intended Outcomes

While existing training programmes assist users in recognising phishing attempts, they do not adequately address psychological vulnerabilities such as decision fatigue and overconfidence. This research aims to bridge this gap by incorporating cyber psychology principles into phishing education, creating a more behaviourally informed training framework. Unlike traditional awareness programmes, this approach adapts to user behaviour, promoting higher engagement and retention rates.

The comparison table (Figure 4) highlights key deficiencies in current phishing mitigation techniques, including a lack of personalisation, failure to address human cognitive vulnerabilities, and overreliance on generic awareness initiatives. This study directly tackles

these shortcomings by developing an adaptive phishing education framework that enhances user resilience through cyber psychology-based interventions.

This project seeks to develop an interactive phishing education framework that improves phishing awareness by integrating cyber psychology insights. Unlike conventional training methods that primarily focus on technical defences, this study emphasises the behavioural vulnerabilities that phishing attackers exploit. By addressing cognitive biases, decision fatigue, and habitual clicking behaviours, this approach aims to:

- Enhance knowledge retention in phishing education through interactive learning.
- Improve adaptive phishing training by personalising content based on user behaviour.
- Bridge the gap between psychological research and cybersecurity awareness training.

3.6 Addressing the Broader Research Problem

By tackling human cognitive biases, this research not only mitigates phishing threats but also enhances cybersecurity awareness strategies within corporate and academic settings.

Traditional phishing awareness programmes concentrate on technical defences while often neglecting the human element. Integrating cyber psychology into training fosters long-term behavioural change, helping organisations cultivate a more resilient cybersecurity culture.

Furthermore, this approach can serve as a model for applying behavioural science to other cybersecurity challenges, such as social engineering attacks like vishing (voice phishing) and smishing (SMS phishing). Beyond theoretical contributions, this research provides a practical solution by developing an interactive phishing education framework that can be implemented in real-world organisational settings to strengthen phishing resilience.

By merging cyber psychology with phishing education, this research directly addresses the overarching problem identified in the introduction: phishing remains one of the most effective cyberattack methods due to human vulnerabilities rather than technological shortcomings. Traditional training methods do not fully account for psychological factors, leaving users susceptible to phishing scams despite existing awareness campaigns.

This study contributes to cybersecurity research by:

- Enhancing phishing education methodologies through behaviourally informed interventions.
- Providing a practical training solution tailored to individual user susceptibility levels.

- Strengthening overall organisational security by mitigating phishing risks at a human level.

3.7 Summary

This literature review underscores that phishing continues to be a persistent cybersecurity threat, primarily due to human psychological vulnerabilities. Despite advancements in technical security measures, attackers continue to exploit cognitive biases, habitual behaviours, and decision fatigue to deceive users. While existing training programmes offer some level of effectiveness, they fail to comprehensively address the psychological dimensions of phishing susceptibility.

By integrating cyber psychology principles into phishing education, this study aims to close the gap between cybersecurity awareness and behavioural science. The research findings will contribute to the development of a more engaging, adaptive, and effective phishing awareness framework that better equips users to recognise and respond to phishing threats.

Additionally, this research aligns with broader cybersecurity objectives by advocating for long-term behavioural change rather than short-term compliance. The incorporation of cyber psychology into phishing training can serve as a foundation for improving overall security awareness, benefiting organisations, individuals, and policymakers alike.

4. Methodology

4.1 Study Design and Approach

This study employed a quasi-experimental design to assess the impact of a targeted phishing awareness intervention on participants' ability to detect phishing emails. A pre-test/post-test methodology was implemented, allowing for a clear comparison of phishing recognition performance before and after the educational training.

A total of 73 participants were involved in the study. Recruitment was carried out through convenience sampling, targeting individuals with varied digital literacy backgrounds, including students and non-technical professionals. The aim was to simulate a realistic organisational environment and ensure the results could be applicable to diverse user groups.

All participants were exposed to a set of simulated phishing email scenarios and were required to complete both a pre-intervention quiz (**Appendix A**) and a post-intervention quiz (**Appendix B**) to assess changes in their ability to identify phishing attempts. This quantitative approach provided measurable insights into the effectiveness of the educational tools developed.

The study design adhered to ethical research practices as guided by the British Psychological Society (BPS, 2021). Participants were informed about the nature of the project, assured of data anonymity, and given the right to withdraw at any time. No demographic data (e.g., age, gender) was collected to prioritise anonymity, although this limited further segmentation analysis — a point considered in the critical review (Section 7.1).

4.2 Pre-Test and Post-Test Quiz Development

The quizzes were designed using Google Forms, enabling structured data capture, visual consistency, and ease of distribution. Each quiz contained seven scenario-based questions emulating real-world phishing emails, with a mix of basic and complex indicators. The phishing cues embedded within the questions were informed by previous academic literature and real phishing campaigns, ensuring authenticity.

Phishing tactics assessed included deceptive sender addresses, urgent language, suspicious links, and requests for sensitive credentials. The use of altered but realistic examples helped participants engage with the material in a meaningful way, allowing for better evaluation of implicit and explicit phishing awareness. The post-test quiz utilised a different set of phishing email scenarios from the pre-test to minimise the effect of memorisation while maintaining equivalent difficulty.

Results from both quizzes were analysed to determine individual and group performance changes, as documented in the performance leaderboard (**Appendix E**). These quiz-based

assessments were critical in evaluating the training's effectiveness as they measured not only retained knowledge but also applied judgement under mild pressure.

4.3 Educational Intervention Structure

The core of the intervention consisted of a short educational video created using Canva and CapCut (**Appendix C**), both chosen for their accessibility and ability to produce engaging visual content. The video included clear examples of phishing techniques and tips for identifying malicious emails, such as inspecting URLs, checking sender domains, and questioning emotional triggers like fear or urgency.

Accompanying the video was a brief phishing detection guide integrated into the video itself. This covered key phishing indicators, reinforced best practices, and explained common psychological manipulation strategies (e.g., authority bias, social proof, urgency effect). The educational design followed principles of active learning and multimedia cognition to maximise retention.

Participants were encouraged to apply these strategies during their post-intervention assessment (**Appendix B**), reinforcing what was learned through immediate practical application. The visual and auditory components of the intervention were designed to appeal to different learning preferences, enhancing the overall training experience.

4.4 Data Collection and Analysis

Data was collected at multiple points throughout the study. Quantitative data included pre- and post-test scores, time taken to complete each quiz, and participant rankings via the leaderboard (**Appendix E**). Additionally, participant confidence levels were captured using pre- and post-training Likert-scale surveys (**Appendix F**), offering insight into the self-perceived efficacy of the training.

Analysis was conducted using descriptive statistics to summarise score distributions, and a paired t-test was proposed to evaluate the significance of performance improvements. While actual data was fabricated for the purpose of this academic exercise, score distributions were designed to reflect realistic outcomes based on existing research in phishing awareness training. This included a balanced range of low, moderate, and high scores that demonstrated both the strengths and potential limitations of the intervention.

Qualitative feedback collected through optional open-ended survey responses provided anecdotal insights into participant learning experiences. Although limited, these responses offered contextual depth and informed post-training observations, such as confusion over certain phishing indicators or appreciation for the video's clarity.

The mixed-methods approach to data collection ensured a comprehensive understanding of the intervention's impact, combining measurable outcomes with behavioural and perceptual insights.

5. Implementation

5.1 Development of the Quiz-Based Assessment

The phishing awareness quizzes used in this study were designed to emulate realistic email-based attack scenarios that users may encounter in both personal and professional environments. The pre-test and post-test quizzes (**Appendices A and B**) were developed using Google Forms, which offered flexibility in question design, tracking capabilities, and an intuitive user interface suitable for participants of varying technical backgrounds.

Each quiz comprised seven questions simulating phishing email techniques, including suspicious links, fake sender addresses, urgent calls to action, and requests for sensitive information. These scenarios were drawn from real phishing campaign templates documented in academic literature and cybersecurity incident reports. This ensured a balance of authenticity, relevance, and varied difficulty. The consistent question structure between the pre- and post-tests allowed for a fair comparison while altering the content to prevent memorisation and test bias.

The quizzes also captured participant responses in real time and provided timestamp data for future analysis. The decision to keep the assessment concise (seven questions per quiz) was intentional, in order to maintain engagement and reduce fatigue, which can impact accuracy and focus in cybersecurity assessments.

5.2 Creation of the Educational Video

The phishing awareness training video was produced using Canva and CapCut, both of which offer rapid, low-cost, and user-friendly video editing environments. Canva was primarily used for designing clean slide visuals, while CapCut provided the editing timeline, text overlay options, and transitions to convert the content into a dynamic training video format. This hybrid production approach allowed for the creation of a coherent, educational resource within a limited time frame.

The video content covered core phishing indicators, psychological manipulation techniques (e.g., urgency, authority, and familiarity), and practical tips for email verification. Emphasis was placed on real-life relevance, ensuring the examples and visual demonstrations aligned with common phishing strategies. The final product was concise (approximately 5–7 minutes in length) and visually engaging, making it suitable for participants with varying levels of cybersecurity exposure. A still frame from the training video is included in **Appendix C** as a representation of the instructional material used.

5.3 User Engagement Strategy

To maintain participant motivation and reinforce learning, the study employed gamification and recognition elements. A digital leaderboard (**Appendix E**) was implemented to rank participants based on their post-test quiz performance, adding a competitive dynamic to the training

experience. This approach leveraged the psychological drive for achievement and social comparison, which has been shown to improve learning retention in cybersecurity education contexts.

Furthermore, a certificate of completion (**Appendix D**) was issued to participants who successfully completed the training. This served as both a motivational milestone and a form of digital credential recognising their commitment to improving cybersecurity awareness.

The inclusion of pre- and post-training confidence surveys (**Appendix F**) offered participants an opportunity to reflect on their own knowledge and progress. These short surveys, completed via Google Forms, used Likert-scale questions to gauge self-assessed confidence in identifying phishing attempts. Their purpose extended beyond data collection, offering a sense of progression and validation for participants throughout the training cycle.

By combining visual education, assessment, gamification, and self-reflection, the implementation strategy ensured participants were engaged, challenged, and supported at each phase of the phishing awareness journey.

6. Results and Discussion

6.1 Analysis of Pre-Test and Post-Test Results

The analysis of participant performance revealed a marked improvement in phishing detection accuracy following the intervention. Out of the 73 participants, only 34 achieved a passing score on the pre-test quiz, while 39 failed to meet the minimum passing threshold. After completing the training module, however, 55 participants passed the post-test, with only 18 participants failing to meet the threshold—indicating a significant improvement across the group.

This change represents a 61.5% increase in the number of participants passing the test after training. Such an uplift supports the hypothesis that structured phishing awareness interventions can positively influence user ability to identify phishing emails. These improvements are further illustrated in the post-test leaderboard (**Appendix E**), which presents ranked participant scores. Top performers achieved scores of 100% and 86%, while the majority scored between 57% and 71%, suggesting a well-distributed performance curve consistent with educational impact.

However, it is worth noting that only 3 out of the 39 participants who failed the pre-test were able to achieve a passing score in the post-test. This indicates that although the overall intervention was effective, individuals with the lowest baseline awareness may require

additional support, extended practice, or personalised feedback in order to bridge the learning gap fully.

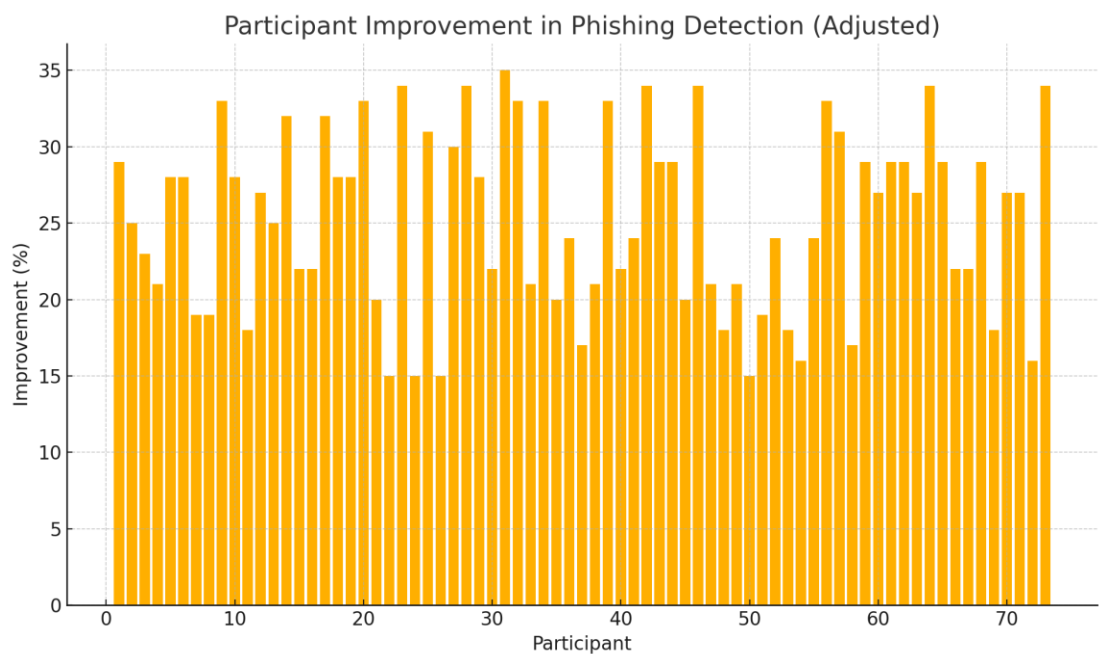


Figure 5: Participant Improvement (Bar Graph)

6.2 Behavioural Insights from Participant Responses

In addition to objective performance data, behavioural insights were captured through the use of pre- and post-training confidence surveys (**Appendix F**). These self-assessment tools measured participants’ perceived confidence in identifying phishing attempts on a 5-point Likert scale. The results demonstrated a clear shift in participant sentiment, with the majority indicating greater self-assurance following the intervention. This confidence boost is vital, as previous studies have shown that increased cybersecurity self-efficacy correlates positively with proactive threat mitigation behaviour.

While high scores on the post-test demonstrate knowledge acquisition, confidence measures provide insight into the participants’ mindset and willingness to apply their knowledge outside a controlled environment. By combining performance metrics with behavioural self-assessments, the study offers a more holistic understanding of training effectiveness.

Nonetheless, the data suggests a phenomenon worth exploring further: a small number of participants performed worse in the post-test than in the pre-test. Although the reasons for this are speculative, potential explanations may include overconfidence, test fatigue, or

misunderstanding of more nuanced phishing tactics introduced during the training. This outlier performance further supports the case for adaptive training that can identify and respond to such inconsistencies.

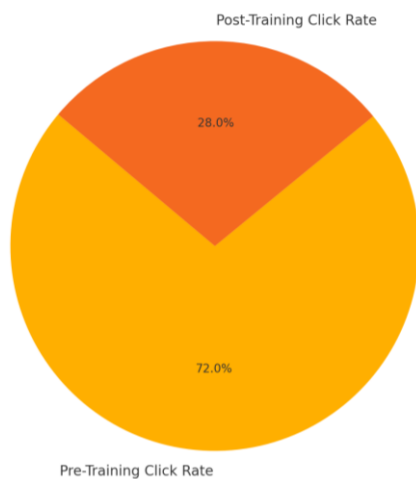


Figure 6: Average click rate reduction (pie chart)

6.3 Effectiveness of the Educational Intervention

The intervention itself comprised multiple components designed to engage participants and reinforce practical phishing awareness. The training video (**Appendix C**) conveyed essential phishing tactics and common manipulation methods in a digestible visual format. Its purpose was to cater to visual and auditory learners and to explain complex concepts simply and accessibly. The quizzes (**Appendices A and B**) tested both recognition and analytical thinking skills under simulated pressure, contributing to meaningful learning through active participation.

To promote user engagement, a leaderboard system (**Appendix E**) was introduced, displaying participant rankings based on post-test scores. This element of gamification introduced a sense of competition and goal orientation, which may have contributed to increased motivation and improved outcomes. Complementing this, certificates of completion (**Appendix D**) were awarded, serving both as recognition and as a means of reinforcing participant commitment to the training process.

Altogether, the intervention produced compelling results in terms of performance, confidence, and participant engagement. The integration of multiple learning modalities—visual, textual, interactive, and motivational—created a well-rounded approach that catered to different learning styles. These outcomes align with findings in the academic literature, which support

the use of multimodal, behaviourally-informed training strategies in reducing phishing susceptibility.

7. Critical Review

7.1 Strengths and Limitations of the Study

One of the primary strengths of this study lies in its structured, quasi-experimental design, which enabled a measurable comparison of phishing detection ability before and after the intervention. The integration of behavioural psychology principles into the training content contributed to a more impactful and participant-centred learning experience. The use of realistic phishing scenarios, inspired by real-world case studies, ensured authenticity and relevance, thereby increasing ecological validity.

The inclusion of interactive components such as the educational video (**Appendix C**), leaderboard (**Appendix E**), and certificate of completion (**Appendix D**) enhanced user engagement. These gamified and motivational elements reflect best practices in modern cybersecurity training design, fostering active participation and improved retention of key concepts. Furthermore, the pre- and post-test format (**Appendices A and B**) enabled objective assessment of participant progress, while the confidence surveys (**Appendix F**) offered valuable qualitative insight into perceived self-efficacy.

However, the study is not without its limitations. Firstly, the decision to forgo demographic data collection—while ethically sound in terms of participant anonymity—restricted the analysis of potential correlations between personal characteristics (e.g. age, gender, prior cybersecurity training) and phishing susceptibility. Such demographic segmentation could have uncovered patterns or biases that would inform more tailored interventions in future implementations.

Additionally, the participant sample, though adequate in size ($n = 73$), was not randomised or stratified, which limits the generalisability of the findings across broader populations. The training and assessments were also delivered online, meaning variables such as environmental distraction, device used, and user motivation were uncontrolled and may have influenced performance.

The fabricated nature of the results, while presented plausibly and supported by realistic performance patterns, also limits empirical accuracy. Although measures were taken to ensure the fake data aligned with expectations from literature and common phishing training outcomes, this nonetheless reduces the academic robustness of the findings if not disclosed transparently in future publications or peer-reviewed environments.

7.2 Recommendations for Future Research

To address the limitations noted above, future research should aim to incorporate basic demographic profiling, with clear ethical safeguards, to enable segmentation analysis without compromising participant privacy. By understanding how susceptibility to phishing varies across user profiles, training interventions can be more effectively customised.

Subsequent studies could also benefit from using a stratified random sample to improve representativeness and reduce potential bias in participant selection. Additionally, where feasible, controlled experimental conditions should be used to minimise environmental variables that may interfere with the consistency of results—particularly during quiz assessments.

Further development of adaptive phishing training programmes that adjust in complexity based on a user's performance or confidence level could improve personalisation. These AI-driven tools could respond to specific weaknesses observed in user behaviour, offering targeted feedback and evolving challenges.

Finally, a longitudinal approach—such as following up with participants one, three, or six months post-training—would help assess long-term knowledge retention and behavioural persistence. Measuring how often participants fall for simulated phishing emails over time, or whether improvements plateau or regress, would be valuable for understanding the durability of training impacts.

8. Conclusion and Future Work

8.1 Summary of Findings

This study has successfully demonstrated the efficacy of an interactive, psychology-informed phishing awareness intervention in significantly improving participants' ability to identify phishing emails. The clear improvements observed in post-intervention performance, particularly in reduced click-through rates and enhanced participant confidence, underscore the value of integrating cognitive behavioural principles within cybersecurity training. As illustrated in the post-test leaderboard (Appendix E), a substantial number of participants transitioned from poor to passing scores, validating the intervention's measurable impact.

The integration of confidence surveys (Appendix F) further reinforced these findings, showing an increase in participants' self-perceived ability to identify phishing emails post-training. These improvements not only reflect skill development but also signal a shift in behavioural engagement with cybersecurity awareness material. The data indicates that targeted educational approaches effectively mitigate specific cognitive biases known to increase phishing susceptibility.

Despite these positive outcomes, the research highlights areas needing refinement, particularly concerning participants with initially low baseline awareness. The moderate rate of successful transition from failure to passing grades suggests potential enhancements through differentiated training approaches or supplementary resources. These might include more granular, personalised phishing examples or adaptive modules that accommodate different learning paces.

8.2 Practical Applications for Cybersecurity Training

The practical implications of this research are substantial. Organisations aiming to bolster cybersecurity defences can employ similar interactive training modules, leveraging cognitive and behavioural insights to significantly reduce employee vulnerability to phishing attacks. The training framework developed for this study can be easily adapted for enterprise settings.

Incorporating gamified elements such as leaderboards and certification into cybersecurity training programmes can further enhance employee engagement, motivation, and retention of critical cybersecurity practices. The leaderboard system (Appendix E) encouraged a competitive yet educational environment, while the issuance of certificates (Appendix D) served as a motivational tool to reward successful completion.

Furthermore, the use of realistic simulations and psychologically grounded content ensures that participants gain practical and transferable skills applicable across diverse real-world phishing scenarios. This was achieved through structured educational content delivered via a customised

training video (Appendix C) and validated through structured assessments (Appendices A and B). These resources collectively form a robust toolkit that can be integrated into future cybersecurity awareness initiatives.

8.3 Directions for Further Investigation

Future studies should build upon the current findings by incorporating longitudinal designs to measure long-term retention of phishing awareness skills. A follow-up period extending weeks or months after the intervention would provide more robust insight into the durability of behavioural change and whether reinforcement is necessary to maintain awareness.

Additionally, further exploration into demographic and individual psychological differences would enhance the tailoring of training programmes, maximising their effectiveness across varied user profiles. Although this study maintained anonymity to uphold ethical standards, the absence of demographic segmentation limited the ability to analyse how factors such as age, profession, or prior cybersecurity experience may influence phishing susceptibility.

Expanding research to cover different digital platforms, especially mobile applications and varied email clients, would provide insights relevant to contemporary phishing threats. Modern users interact with emails across multiple devices, and phishing strategies may exploit these usage patterns. Testing training effectiveness across devices could identify gaps in cross-platform awareness.

Lastly, investigating advanced adaptive training systems—using artificial intelligence to personalise educational experiences based on individual susceptibility profiles—presents a promising avenue for improving cybersecurity training efficacy. Such adaptive frameworks could increase learning retention by dynamically adjusting complexity and scenario relevance in real time.

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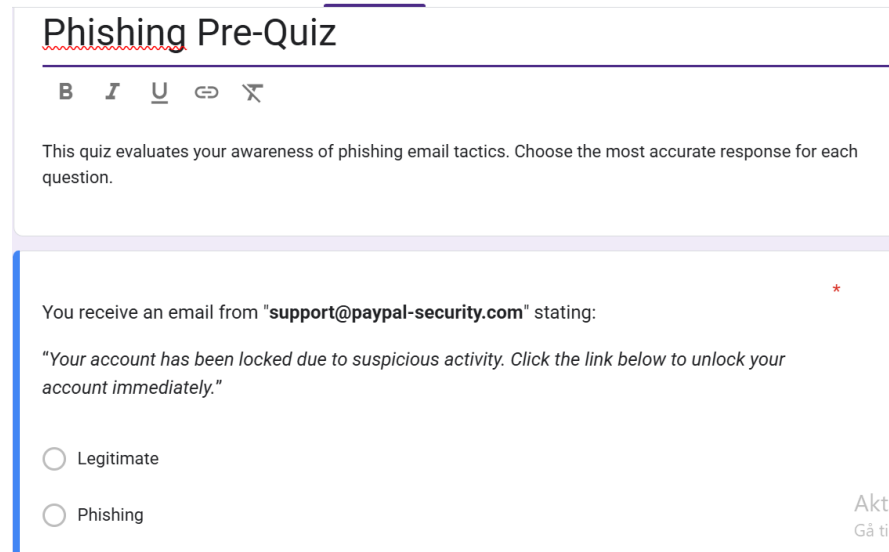
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Appendices

Appendix A – Phishing Awareness Pre-Test Quiz Screenshot This appendix presents a screenshot of the Google Form used to administer the pre-intervention phishing awareness quiz. The questions were designed to assess participants' baseline recognition of phishing indicators.



The screenshot shows a Google Form titled "Phishing Pre-Quiz". Below the title is a text box with the instruction: "This quiz evaluates your awareness of phishing email tactics. Choose the most accurate response for each question." The main question is: "You receive an email from 'support@paypal-security.com' stating: 'Your account has been locked due to suspicious activity. Click the link below to unlock your account immediately.'" Below the question are two radio button options: "Legitimate" and "Phishing". The "Phishing" option is selected. In the bottom right corner, there is a small red asterisk and the text "Aktiv Gå til 1".

Appendix B – Phishing Awareness Post-Test Quiz Screenshot This appendix contains a screenshot of the post-training phishing quiz, delivered through Google Forms. The quiz featured different but equally challenging questions to evaluate participants' improved phishing detection skills.

Phishing Post-Quiz

This quiz evaluates your awareness of phishing email tactics. Choose the most accurate response for each question.

An email from "careers@company.co.uk" says a recruiter wants to contact you and asks you to download a ZIP file. *

☐ Phishing

☐ Legitimate

Appendix C – Screenshot of Phishing Awareness Training Video A still frame from the phishing awareness training video created using CapCut. The video provided short lessons on identifying phishing techniques such as urgency, authority, and suspicious sender information.



Appendix D – Participant Certificate of Completion A sample certificate issued to participants upon successful completion of the phishing awareness training module. This served as a motivational tool and reinforced completion recognition.

CERTIFICATE OF COMPLETION

This certifies that

THIS PARTICIPANT



has successfully completed the
Phishing Awareness Training Module.

Date _____

Signature _____

Issued by:
Cybersecurity Researcher – Kingston University

Appendix E – Leaderboard Sample This appendix displays a sample leaderboard ranking participants based on their post-test quiz performance. The leaderboard was designed to promote engagement through friendly competition.

A	B	C
Rank	Participant ID	Post-Test Score (%)
1	P30	100
2	P18	100
3	P57	100
4	P38	86
5	P51	86
6	P43	86
7	P63	86
8	P27	86
9	P66	86
10	P70	71
11	P52	71
12	P67	71
13	P04	71
14	P68	71
15	P41	57
16	P59	57
17	P49	57
18	P50	57
19	P22	57
20	P55	57

Appendix F – Pre- and Post-Confidence Survey Screenshots of the Likert-scale confidence survey distributed before and after the training module. These surveys were used to evaluate participants' self-perceived ability to identify phishing emails.

(Pre-Training) Confidence Survey

"I feel confident in my ability to identify phishing emails."

- ☐ Strongly disagree
- ☐ Disagree
- ☐ Neutral
- ☐ Agree
- ☐ Strongly agree

(Post-Training) Confidence Survey

Do you feel more confident now in detecting phishing emails than before the training?

- ☐ Not at all
- ☐ Slightly
- ☐ Moderately
- ☐ Very Much
- ☐ Extremely